

SONAL BHATIA

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Education

Princeton University

January 2027

B.S.E. Mechanical & Aerospace Engineering, Minors in Computer Science and Robotics

Princeton, NJ

- **Relevant Coursework:** Algorithms & Data Structures, Computer Architecture, Computer Vision, Operating Systems, Distributed Systems, Microprocessors, Robotics, Ethics of Computing, Optimal Control, High Tech Entrepreneurship
- **Extracurricular:** Rocketry Club, Autonomous Vehicles Engineering, Society of Women Engineers, Rewriting the Code

Experience

Apple Inc.

June – August 2026

Software Engineer Intern, AI Performance & Systems

Cupertino, CA

- **Built and shipped fixes** for 10 issues across graphics and ML **inference pipelines**, improving reliability and performance on next-generation MacBook Pro and Mac Studio **system and SoC architectures**.
- Diagnosed a compute-bound bottleneck in a production **LLM MoE inference** pipeline; **built internal profiling tooling** that surfaced 3 additional issues, boosting performance by 30% and driving engineering decisions through stakeholder presentations.

Apple Inc.

January – August 2025

Software Engineer Intern, Applied AI & Systems

Beaverton, OR

- **Engineered a 3D-GenAI post-processing pipeline from scratch**, implementing Dual Contouring, Catmull-Clark, and OpenVDB algorithms as production modules, enabling on-device 3D quad-mesh generation up to **1024³** resolution.
- **Built and deployed a model quantization toolkit** (INT8, INT4, mixed-precision) across 8 GenAI models, cutting memory footprint by **44%** and unlocking on-device deployment; instrumented inference pipelines to raise GPU utilization by **15%**.

The Estée Lauder Companies Inc.

August – December 2024

Software Engineer Intern, Data Science

New York City, NY

- **Designed and shipped a Bayesian Regression model** trained on 50,000+ transactions into a targeting pipeline, improving accuracy by **40%**; built a **Monte Carlo/MCMC** inference pipeline for a Regression Discontinuity Design, reducing confidence interval widths by **15%**.

Johns Hopkins University Applied Physics Laboratory

June – October 2024

Software Development Engineer Intern

Laurel, MD

- Built classified computer graphics software in **C++** for threat signature modeling; authored 75+ automated test cases, resolving cross-API rendering discrepancies between **OpenGL and Vulkan** backends by **80%**.

Princeton University

Spring & Fall 2024, Spring 2026

Undergraduate Course Assistant – Linear Systems (EGR 154) & Intro to Computer Science

Princeton, NJ

- Led help sessions for **250+ students** across 3 semesters, teaching **linear algebra**, systems theory, and **CS fundamentals**.

EMTech Space S.A.

June – August 2023

Software Engineer Intern, Mission Design

Athens, Greece

- Designed and implemented a **genetic algorithm**-based orbit design tool, cutting computation time by **30%** across 1,000+ scenarios; authored 3 technical reports approved by the European Space Agency for the Hellenic Space Dawn CubeSat mission.

Leadership

Break Through Tech AI, Cornell University

May 2024 – April 2025

Software Engineering & Machine Learning Fellow

Remote

- Built ML pipelines improving model accuracy by **18%** and reducing log loss by **12%** through hyperparameter tuning and regularization; obtained certificate in ML foundations at Cornell University.

Princeton Rocketry Club

September 2022 – December 2024

Co-President; prev. Systems Engineer, Team Lead, & Professional Development Officer

Princeton University

- Managed 7 cross-functional project teams across **95+ members**; planned a student-led initiative to design and launch a two-stage solid-motor rocket targeting suborbital altitude.

Skills

Languages: Python, Java, C++, Go, Swift, Objective-C, Assembly, HTML, CSS, JavaScript, TypeScript

Systems & APIs: REST APIs, Node.js, React, SQL/NoSQL databases, Docker, Git, CI/CD, AWS/GCP

AI/ML: PyTorch, TensorFlow, MLX, Core ML, Metal Performance Shaders, CUDA, LLM tooling, OpenCV

Quantitative: Linear Algebra, Bayesian Regression, Monte Carlo/MCMC, Kalman Filtering

Projects

GPU Acceleration of Shap-E via PyTorch MPS | *Python, Metal Performance Shaders*

March – July 2025

- **Built and shipped a GPU-acceleration patch** for OpenAI's Shap-E repository, enabling image-to-3D **inference** on Apple Silicon and cutting inference time from **4+ hours to under 4 minutes**; filed as a [Pull Request](#), rewriting core tensor-indexing logic while preserving CUDA/CPU compatibility.

Autonomous Obstacle-Avoiding Drone | *Python, OpenCV, CrazyFlie, Kalman Filter*

September – December 2024

- Built a fully autonomous CrazyFlie drone with **real-time HSV obstacle detection**, **contour-based avoidance**, and **closed-loop waypoint following** via an onboard Kalman filter.